SQL END-TO-END
PROJECT
Data Extraction | Transformation | Visualization | Reporting
SELECT *
FROM data_table;

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New Query

retail_sales Execute

Object Explorer

Connect

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- XEvent Profiler

sql_script.s...\BEGON (88))

```sql
-- FINAL RETAIL SALES ANALYTICS PROJECT (SQL SERVER VERSION)

/* ============================================================
    1. PROJECT OVERVIEW
   Fully documented SQL Server Retail Analytics project including:
   - Database creation
   - Schema design
   - CSV import using BULK INSERT
   - Data cleaning
   - KPI Analysis
   - Views
   - Stored Procedures
   - Functions
   - Window Functions
   - CTEs
============================================================ */

/* ============================================================
    2. CREATE DATABASE
============================================================ */
IF NOT EXISTS (SELECT name FROM sys.databases WHERE name='retail_sales')
BEGIN
    CREATE DATABASE retail_sales;
END;
GO
USE retail_sales;
GO

/* ============================================================
    3. CREATE TABLES
============================================================ */
IF OBJECT_ID('customers') IS NOT NULL DROP TABLE customers;
IF OBJECT_ID('products') IS NOT NULL DROP TABLE products;
IF OBJECT_ID('orders') IS NOT NULL DROP TABLE orders;
IF OBJECT_ID('order_items') IS NOT NULL DROP TABLE order_items;
GO

CREATE TABLE customers (
```

100 % 2 0 Ln: 1, Ch: 1 SPC LF UTF-8

Connected. (1/1) (local)\SQLEXPRESS (17.0 RTM) BEAST\BEGON (88) retail_sales 00:00:00 Row: 0, Col: 0 0 rows

Ready Select Repository

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New Query

retail_sales Execute

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sql_script.s...\BEGON (88))

```sql
        3. CREATE TABLES
    ============================================================ */
    IF OBJECT_ID('customers') IS NOT NULL DROP TABLE customers;
    IF OBJECT_ID('products') IS NOT NULL DROP TABLE products;
    IF OBJECT_ID('orders') IS NOT NULL DROP TABLE orders;
    IF OBJECT_ID('order_items') IS NOT NULL DROP TABLE order_items;
    GO

    CREATE TABLE customers (
        customer_id INT PRIMARY KEY,
        customer_name NVARCHAR(100),
        region NVARCHAR(50),
        signup_date DATE
    );

    CREATE TABLE products (
        product_id INT PRIMARY KEY,
        product_name NVARCHAR(100),
        category NVARCHAR(50),
        price DECIMAL(10,2)
    );

    CREATE TABLE orders (
        order_id INT PRIMARY KEY,
        customer_id INT,
        order_date DATE,
        order_status NVARCHAR(20),
        FOREIGN KEY (customer_id) REFERENCES customers(customer_id)
    );

    CREATE TABLE order_items (
        order_item_id INT PRIMARY KEY,
        order_id INT,
        product_id INT,
        quantity INT,
        unit_price DECIMAL(10,2),
        FOREIGN KEY (order_id) REFERENCES orders(order_id),
        FOREIGN KEY (product_id) REFERENCES products(product_id)
    );
```

100 % 2 0 Ln: 1, Ch: 1 SPC LF UTF-8

Connected. (1/1) (local)\SQLEXPRESS (17.0 RTM) BEAST\BEGON (88) retail_sales 00:00:00 Row: 0, Col: 0 0 rows

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New Query

retail_sales Execute

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sql_script.s...\BEGON (88))

```sql
GO

/* ============================================================
    4. CSV IMPORT USING BULK INSERT
============================================================ */
-- NOTE: SQL Server requires permissions for BULK INSERT
-- Right click folder → Properties → Security → Add → NT SERVICE\MSSQLSERVER → Read

BULK INSERT customers
FROM 'C:\Users\BEGON\Desktop\SQL\projects\retail_FINAL\raw/customers.csv'
WITH (
    FORMAT='CSV',
    FIRSTROW = 2,
    FIELDTERMINATOR = ',',
    ROWTERMINATOR = '/n/s'
);
GO

BULK INSERT products
FROM 'C:\Users\BEGON\Desktop\SQL\projects\retail_FINAL\raw/products.csv'
WITH (
    FORMAT='CSV',
    FIRSTROW = 2,
    FIELDTERMINATOR = ',',
    ROWTERMINATOR = ''
);
GO
BULK INSERT orders
FROM 'C:\Users\BEGON\Desktop\SQL\projects\retail_FINAL\raw/orders.csv'
WITH (
    FORMAT='CSV',
    FIRSTROW = 2,
    FIELDTERMINATOR = ',',
    ROWTERMINATOR = ''
);
GO

BULK INSERT order_items
FROM 'C:\Users\BEGON\Desktop\SQL\projects\retail_FINAL\raw/order_items.csv'
```

100 % 2 0 Ln: 1, Ch: 1 SPC LF UTF-8

Connected. (1/1) (local)\SQLEXPRESS (17.0 RTM) BEAST\BEGON (88) retail_sales 00:00:00 Row: 0, Col: 0 0 rows

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New Query

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sql_script....\BEGON (88))*

```
        FORMAT='CSV',
        FIRSTROW = 2,
        FIELDTERMINATOR = ',',
        ROWTERMINATOR = ''
    );
    GO

    Select * from customers
    Select * from products
    Select * from order_items
    Select * from orders
    /* ============================================================
        5. DATA CLEANING
    ============================================================ */
    SELECT * FROM customers WHERE region IS NULL;
    SELECT * FROM products WHERE price IS NULL;
    SELECT * FROM orders WHERE order_date IS NULL;
    SELECT category FROM products
    group by category
```

100 % 2 0 Ln: 116, Ch: 1 (94 chars, 5 lines) SPC LF UTF-8

Results Messages

	customer_id	customer_name	region	signup_date
1	1	Customer_1	West	2023-02-14
2	2	Customer_2	East	2023-04-08
3	3	Customer_3	West	2020-11-26
4	4	Customer_4	East	2021-09-25
5	5	Customer_5	South	2022-03-03

	product_id	product_name	category	price
1	1	Product_1	Sports	742.33
2	2	Product_2	Grocery	332.11
3	3	Product_3	Electronics	761.36
4	4	Product_4	Grocery	653.09
5	5	Product_5	Home	69.88

	order_item_id	order_id	product_id	quantity	unit_price
4	4	2866	150	1	938.76
5	5	2623	123	1	122.97
6	6	96	78	1	811.64
7	7	749	20	2	690.21

	order_id	customer_id	order_date	order_status
19	19	332	2023-06-01	Completed

Query executed successfully. (local)\SQLEXPRESS (17.0 RTM) BEAST\BEGON (88) retail_sales 00:00:00 Row: 1, Col: 1 10,200 rows

Ready Select Repository

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New Query

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sql_script....\BEGON (88))*

```
Select * from order_items
Select * from orders
/* ============================================================
    5. DATA CLEANING
============================================================ */
SELECT * FROM customers WHERE region IS NULL;
SELECT * FROM products WHERE price IS NULL;
SELECT * FROM orders WHERE order_date IS NULL;
SELECT category FROM products
group by category
UPDATE products SET category='Electronics' WHERE category IN ('electronics','Elec','ELECTRO');
UPDATE order_items SET quantity=1 WHERE quantity <= 0;
GO

select  top 2 COUNT(product_id)as total,category from products
group by category
order by total desc

/* ============================================================
```

100 % 2 0 Ln: 127, Ch: 1 (137 chars, 3 lines) SPC LF UTF-8

Results Messages

customer_id	customer_name	region	signup_date

Click to select the whole column

product_id	product_name	category	price

order_id	customer_id	order_date	order_status

Query executed successfully. (local)\SQLEXPRESS (17.0 RTM) BEAST\BEGON (88) retail_sales 00:00:00 Row: 1, Col: 1 0 rows

Ready Select Repository

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New Query

retail_sales Execute

Object Explorer

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sql_script....\BEGON (88))*

```sql
Select * from order_items
Select * from orders
/* ============================================================
    5. DATA CLEANING
============================================================ */
SELECT * FROM customers WHERE region IS NULL;
SELECT * FROM products WHERE price IS NULL;
SELECT * FROM orders WHERE order_date IS NULL;
SELECT category FROM products
group by category
UPDATE products SET category='Electronics' WHERE category IN ('electronics','Elec','ELECTRO');
UPDATE order_items SET quantity=1 WHERE quantity <= 0;
GO

select  top 2 COUNT(product_id)as total,category from products
group by category
order by total desc

/* ============================================================
```

100 % 2 0 Ln: 131, Ch: 1 (150 chars, 2 lines) SPC LF UTF-8

Messages

```
(26 rows affected)

(0 rows affected)

Completion time: 2026-04-02T16:08:37.9683791+05:30
```

100 % No issues found Ln: 7, Ch: 1 TABS MIXED UTF-8 with BOM

Query executed successfully. (local)\SQLEXPRESS (17.0 RTM) BEAST\BEGON (88) retail_sales 00:00:00 Row: 1, Col: 1 0 rows

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New Query

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sql_script....\BEGON (88))*

```sql
SELECT SUM(quantity * unit_price) AS total_revenue FROM order_items;
GO

-- Monthly Revenue
SELECT FORMAT(order_date,'yyyy-MM') AS month,
       SUM(quantity * unit_price) AS revenue
FROM orders o JOIN order_items oi ON o.order_id=oi.order_id
WHERE order_status='Completed'
GROUP BY FORMAT(order_date,'yyyy-MM')
ORDER BY month;
GO

-- Top Selling Products
SELECT P.category,p.product_name, SUM(oi.quantity) AS total_units_sold
FROM order_items oi JOIN products p ON oi.product_id=p.product_id
GROUP BY p.product_name,p.category
ORDER BY total_units_sold DESC;
GO
```

100 % 2 0 Ln: 159, Ch: 1 (563 chars, 18 lines) SPC LF UTF-8

Results Messages

	total_revenue
1	7579554.15

	month	revenue
1	2021-01	145970.02
2	2021-02	107274.92
3	2021-03	197026.31
4	2021-04	122437.10
5	2021-05	142744.23
6	2021-06	116354.48
7	2021-07	146949.16
8	2021-08	189480.57

	category	product_name	total_units_sold
1	Fashion	Product_21	117
2	Grocery	Product_160	112
3	Home	Product_129	107
4	Fashion	Product_181	107
5	Home	Product_23	104
6	Sports	Product_7	104

Query executed successfully. (local)\SQLEXPRESS (17.0 RTM) BEAST\BEGON (88) retail_sales 00:00:00 Row: 1, Col: 1 241 rows

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New Query

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sql_script....\BEGON (88))*

```sql
GO

/* ============================================================
    7. ADVANCED SQL FEATURES (DOCUMENTED)
============================================================ */
SELECT * FROM order_items
SELECT * FROM products

select p.category,SUM(quantity*unit_price ) as total_sales from products as p
join order_items as o on p.product_id = o.order_id
group by category
order by total_sales desc


/*  VIEW: Monthly Revenue */
CREATE OR ALTER VIEW vw_monthly_revenue AS
SELECT FORMAT(order_date,'yyyy-MM') AS month,
       SUM(quantity * unit_price) AS monthly_revenue
FROM orders o JOIN order_items oi ON o.order_id=oi.order_id
```

100 % 2 0 Ln: 170, Ch: 1 (173 chars, 4 lines) SPC LF UTF-8

Results Messages

	category	total_sales
1	Fashion	137920.37
2	Home	119657.15
3	Grocery	118813.95
4	Sports	93303.01
5	Electronics	71514.61

Query executed successfully. (local)\SQLEXPRESS (17.0 RTM) BEAST\BEGON (88) retail_sales 00:00:00 Row: 1, Col: 1 5 rows

Ready Select Repository

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New Query

retail_sales Execute

Object Explorer

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sql_script....\BEGON (88))*

```
join order_items as o on p.product_id = o.order_id
group by category
order by total_sales desc


/*  VIEW: Monthly Revenue */
CREATE OR ALTER VIEW vw_monthly_revenue AS
SELECT FORMAT(order_date,'yyyy-MM') AS month,
       SUM(quantity * unit_price) AS monthly_revenue
FROM orders o JOIN order_items oi ON o.order_id=oi.order_id
WHERE order_status='Completed'
GROUP BY FORMAT(order_date,'yyyy-MM');
GO
select *  from vw_monthly_revenue

Go

/*  STORED PROCEDURE: Get CLV */
CREATE OR ALTER PROCEDURE get_clv @cust_id INT AS
```

100 % 2 0 Ln: 180, Ch: 1 (34 chars, 1 lines) SPC LF UTF-8

Results Messages

	month	monthly_revenue
1	2021-01	145970.02
2	2021-02	107274.92
3	2021-03	197026.31
4	2021-04	122437.10
5	2021-05	142744.23
6	2021-06	116354.48
7	2021-07	146949.16
8	2021-08	189480.57
9	2021-09	142733.50
10	2021-10	169238.80
11	2021-11	148091.41
12	2021-12	105383.74
13	2022-01	153431.18
14	2022-02	143077.84
15	2022-03	187472.58
16	2022-04	130184.77
17	2022-05	138510.90
18	2022-06	167866.82

Query executed successfully. (local)\SQLEXPRESS (17.0 RTM) BEAST\BEGON (88) retail_sales 00:00:00 Row: 1, Col: 1 40 rows

Ready Select Repository

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New Query

retail_sales Execute

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sql_script....\BEGON (88))*

```sql
Go

/*  STORED PROCEDURE: Get CLV */
CREATE OR ALTER PROCEDURE get_clv @cust_id INT AS
BEGIN
    SELECT c.customer_id, c.customer_name,
           SUM(oi.quantity * oi.unit_price) AS lifetime_value
    FROM customers c
    JOIN orders o ON c.customer_id=o.customer_id
    JOIN order_items oi ON o.order_id=oi.order_id
    WHERE c.customer_id=@cust_id AND o.order_status='Completed'
    GROUP BY c.customer_id, c.customer_name;
END;
GO
exec get_clv @cust_id = 2

/*  FUNCTION: Order Value */

CREATE OR ALTER FUNCTION order_value(@orderId INT)
```

100 % 2 0 Ln: 196, Ch: 1 (26 chars, 1 lines) SPC LF UTF-8

Results Messages

	customer_id	customer_name	lifetime_value
1	2	Customer_2	1320.35

Query executed successfully. (local)\SQLEXPRESS (17.0 RTM) BEAST\BEGON (88) retail_sales 00:00:00 Row: 1, Col: 1 1 rows

Ready Select Repository

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New Query

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sql_script....\BEGON (88))*

```sql
/*  FUNCTION: Order Value */

CREATE OR ALTER FUNCTION order_value(@orderId INT)
RETURNS DECIMAL(10,2)
AS
BEGIN
    DECLARE @total DECIMAL(10,2);
    SELECT @total = SUM(quantity*unit_price)
    FROM order_items WHERE order_id=@orderId;
    RETURN @total;
END;
GO
/*  WINDOW FUNCTIONS */
-- ROW_NUMBER
SELECT customer_id, customer_name, region,
 ROW_NUMBER() OVER(PARTITION BY region ORDER BY signup_date) AS region_rank
FROM customers;
GO

-- DENSE_RANK
SELECT product_id, SUM(quantity) AS total_qty,
 DENSE_RANK() OVER(ORDER BY SUM(quantity) DESC) AS rank_position
FROM order_items
GROUP BY product_id;
GO

-- NTILE
SELECT customer_id,
 SUM(quantity*unit_price) AS revenue,
 NTILE(100) OVER(ORDER BY SUM(quantity*unit_price) DESC) AS centile
FROM orders o JOIN order_items oi ON o.order_id=oi.order_id
WHERE order_status='Completed'
GROUP BY customer_id;
GO

/*  CTE */
WITH customer_sales AS (
 SELECT c.customer_id, c.customer_name, SUM(oi.quantity*oi.unit_price) AS total_spent
```

100 % 2 0 Ln: 198, Ch: 29 SPC LF UTF-8

Query executed successfully. (local)\SQLEXPRESS (17.0 RTM) BEAST\BEGON (88) retail_sales 00:00:00 Row: 1, Col: 1 1 rows

Ready Select Repository

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New Query

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sql_script....\BEGON (88))*

```sql
SELECT customer_id, customer_name, region,
 ROW_NUMBER() OVER(PARTITION BY region ORDER BY signup_date) AS region_rank
FROM customers;
GO

-- DENSE_RANK
SELECT product_id, SUM(quantity) AS total_qty,
 DENSE_RANK() OVER(ORDER BY SUM(quantity) DESC) AS rank_position
FROM order_items
GROUP BY product_id;
GO

-- NTILE
SELECT customer_id,
 SUM(quantity*unit_price) AS revenue,
 NTILE(100) OVER(ORDER BY SUM(quantity*unit_price) DESC) AS centile
FROM orders o JOIN order_items oi ON o.order_id=oi.order_id
WHERE order_status='Completed'
GROUP BY customer_id;
```

100 % 2 0

2 errors
Click to navigate to the Error List

Ln: 225, Ch: 1 (241 chars, 7 lines) SPC LF UTF-8

Results Messag

	customer_id	revenue	centile
1	182	33232.78	1
2	466	29464.78	1
3	483	27875.90	1
4	936	27743.56	1
5	179	27613.73	1
6	451	25140.36	1
7	707	24781.94	1
8	526	23704.09	1
9	999	23249.44	1
10	753	22274.76	2
11	808	22255.37	2
12	712	22163.83	2
13	965	21958.51	2
14	902	21539.79	2
15	14	20939.17	2
16	570	20876.60	2
17	991	20631.71	2
18	639	20058.21	2

Query executed successfully. (local)\SQLEXPRESS (17.0 RTM) BEAST\BEGON (88) retail_sales 00:00:00 Row: 1, Col: 1 875 rows

Ready Select Repository